Same Selection, Opposite Rankings

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Abstract

A favorable selection result can depend on which admissible reference is used and how that reference responds to computational support. We first show that replacing one control matched on source, available task metadata, and token count with another reverses a fixed selected dataset's point ranking under both Qwen training bundles, moving its reported advantage by 6.57 and 5.49 percentage points. We then cross two references with replay and execution on the same Mistral GSM8K cohort. Supplying actual results at paired generated prefixes reduces the selected-minus-A gap by 12.89 points, from $+10.94$ to -1.95, but reduces the selected-minus-B gap by only 3.13 points. The difference between these changes is 9.77 points, with a conditional 95% interval of [4.69,14.84]. Thus, the reference determines both the estimated advantage and how much that advantage changes under execution. Repair–harm accounting, two fixed fits per arm, a separate question cohort, and answer-review sensitivity support the focal diagnostic. These retrospectively selected comparisons show why balance-certified references cannot be presumed interchangeable when interpreting a selection advantage.

1 Introduction

Suppose a model trained on selected instruction data beats a carefully matched random baseline. Has the selection method identified better data? The natural reading is that, once obvious composition and budget differences are removed, the remaining performance gap measures the value of the selected examples. We test the stability of that reading by asking: *can the estimated ordering reverse without changing the selected training set?*

This question starts from a distinction between a selection signal, the composition it induces, and downstream utility. A likelihood score ranks responses under a scorer; it does not directly measure the performance of a model subsequently trained on those responses. Length and style can themselves be useful, while also influencing the score (Zhao et al., 2024; Li et al., 2025b; Wang et al., 2026a). A selected set may therefore win because of its mixture of examples, their within-group membership, or the way the resulting model meets an evaluation interface. Separating these possibilities requires specifying what the comparison holds fixed.

Our starting research question asks whether an advantage remains after reproducing the selected set's observed composition in a random reference. A positive residual is useful evidence, but only relative to that realized reference. We take the next step: retain the selected set and examine whether its reported advantage survives replacing the matched reference or supplying a computational result at evaluation. This turns a broad question about better data into two concrete tests of the stability of a claimed benefit.

The point ordering can reverse, and the two tests form a diagnostic sequence (Figure 1). A fixed selected set ranks above matched control A but below matched control B under both Qwen training bundles. Both controls satisfy the same composition and token constraints; each selected fit is shared between the comparisons. We then examine whether these references are interchangeable in their response to computational support. On the same Mistral GSM8K questions, paired execution reduces the selected-minus-A gap by $12.89\,\mathrm{pp}$ but the selected-minus-B gap by $3.13\,\mathrm{pp}$. Their difference, $9.77\,\mathrm{pp}$, directly connects comparator choice with support dependence. The first comparison establishes the reference problem; the second characterizes it through a local intervention.

Prior work already documents evaluation-dependent selection trends and interface-dependent reversals of measured training effects (Diddee and Ippolito, 2025; Gan et al., 2026). Our focus is more specific: we hold the selected fit fixed while replac-

ing an admissible matched reference, and then hold each generated program/prefix fixed while replacing its computational feedback. The contribution is the evidence from these controlled comparisons, rather than the general observation that evaluation conditions matter.

Compact instruction datasets can support strong adaptation (Zhou et al., 2023a). Selection methods exploit likelihood, quality, diversity, or influence (Albalak et al., 2024; Qin et al., 2024; Liu et al., 2024; Xia et al., 2024), with Superfiltering and T-SHIRT providing inexpensive likelihood-based approaches (Li et al., 2024a; Fu et al., 2025). We examine how an apparent benefit from an SF-style selected subset should be interpreted after constructing composition-matched references.

We make three connected empirical contributions. First, we cross two accepted references with two training bundles and find reference-induced shifts larger than the selected set's original advantage. Second, we cross those reference memberships with replay and execution on a common Mistral cohort, showing that the change in selection advantage itself depends on the comparator. Third, we account for the focal change through repairs and harms and examine it across fixed fits, disjoint questions, and answer interpretation. The matched-reference comparison motivates the diagnostic; the within-cohort execution comparison supplies the connecting evidence.

2 One Contrast, Two Interventions

Let M_S be a model trained on selected set S, M_C a model trained on control C, and $q(M, e)$ its score under evaluation interface e. For a fixed benchmark and training bundle, define

$$G(S, C, e) = q(M_S, e) - q(M_C, e). \quad (1)$$

A positive value favors the selected model. Reading this value as a stable advantage of S requires distinguishing the selected set from the other arguments in the comparison.

The reference defines the comparison. An unmatched random reference compares the selection pipeline with parent-pool sampling. A composition-matched reference instead asks whether the selected examples outperform alternatives with specified observed features. Our controls A and B satisfy the same source, observed-task, and token constraints, denoted M1 (Section 3.1). A positive gap against one realization need not persist against another.

Intervention 1: replace the matched comparator. Hold S, its fitted model, and e fixed; replace $C = A$ with $C = B$ under the same matching specification. The resulting change is

$$G(S, B, e) - G(S, A, e) = q(M_A, e) - q(M_B, e). \quad (2)$$

Because the selected fit cancels, the change isolates how the reported advantage depends on the realized reference. If the sign reverses, a positive matched residual cannot define a comparator-independent ranking of the selected data.

Intervention 2: cross references with execution. Hold the trained models fixed and compare replay (e_r) with execution (e_x) at paired generated prefixes. Let $\Delta_g = q(M_g, e_x) - q(M_g, e_r)$ and, for each comparator C, define the reduction in the selection gap as

$$I_C = G(S, C, e_r) - G(S, C, e_x) = \Delta_C - \Delta_S. \quad (3)$$

The common-cohort comparison then gives

$$H = I_A - I_B = \Delta_A - \Delta_B. \quad (4)$$

This difference asks whether changing the comparator changes the inferred sensitivity of selection advantage to execution. The selected-model term cancels, so H is also a control–control response contrast. It cannot identify a selection-specific training mechanism by itself. We report it alongside G and I_A, which retain the selected model, to connect reference choice with the measured advantage.

These identities organize measured contrasts, rather than provide a new theorem. Reference replacement tests the initial comparison; paired execution diagnoses how the comparison changes when actual computational feedback is supplied. The Mistral crossing evaluates their connection within one family and question cohort.

3 Controlled Comparisons

3.1 Data, Controls, and Training

We study 1,000-example instruction subsets from a frozen 100,000-example pool derived from the training split of `allenai/tulu-3-sft-mixture` (Lambert et al., 2025). The focal reservation contains 20,000 examples, of which 19,791 pass scorer

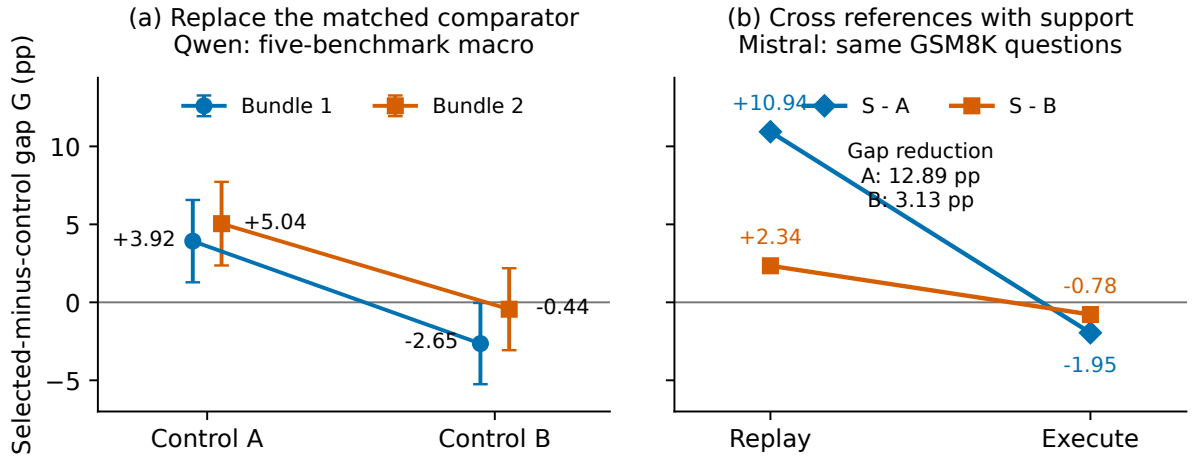


Figure 1: **Reference sensitivity motivates a paired execution diagnostic.** Left: Qwen selected-minus-control macro scores under comparator replacement, with approximate 95% paired, stratified normal intervals (Appendix I.3). Right: Mistral $S - A$ and $S - B$ gaps on the same GSM8K cohort under replay and execution. The right panel shows point estimates; uncertainty for their changes is reported in the text. The panels use different model families and outcomes, while the two right-panel lines share models, questions, and support conditions.

eligibility. The selected set uses a Superfiltering-style GPT-2 conditional/unconditional response-perplexity ratio, taking the highest eligible ratios strictly below one (Radford et al., 2019; Li et al., 2024b,a). Appendix F specifies the dataset revision, reservation procedure, realized category counts, and outcome-informed history of the focal A/B analysis.

Controls A/B match declared source and observed task-family categories under the same M1 acceptance rules (Table 1). Source counts are fixed by construction. Only 45 examples per set have task-family labels; the other 955 are UNKNOWN. M1 therefore balances sparse metadata, not comprehensive semantic task types. Each proposal has 1,000 examples and a single-pass target-token total within 2% of S, measured under the Qwen rendering/tokenizer used for control construction. The same example IDs are subsequently reused for Mistral; token matching is not redefined as identical tokenization across families. Categorical checks bound Jensen–Shannon divergence and total variation and require coverage of the selected set's categorical mass.

The sampler excludes S by example, group, and duplicate identity. It uses calibrated proposals and accepts the first feasible subset among at most 64 attempts. This is constrained sampling, not uniform sampling over all feasible subsets. Controls may overlap one another: selected–control disjointness does not imply mutual disjointness of every control. The realized controls have exactly the selected

Constraint shared by A/B	Requirement
Examples per subset	1,000
Observed categorical features	Source; task family
Qwen-rendered target-token total	Relative error $\leq 2\%$
Categorical JS / TV	$\leq .01$ / $\leq .02$
Categorical mass coverage	$\geq .99$
Selected-set exclusion	Example/group/duplicate
Proposal acceptance	First feasible of ≤ 64 proposals

Table 1: **The same matching specification, different realized controls.** These are acceptance criteria, not claims of identical semantic content. JS denotes Jensen–Shannon divergence and TV total variation.

set's source counts and task-family total variation of .001 each. Their single-pass target-token totals are 582,713 and 581,067, versus 592,395 for S: errors of 1.63% and 1.91%. Thus, the reported reversal occurs between two accepted controls, not between a balanced control and a rejected proposal. Appendix B records their saved checks. These details define the reference distribution against which a residual is measured (Stuart, 2010).

In the crossed Qwen comparison, the same $S/A/B$ memberships are evaluated under two random-number-generator (RNG) bundles. Within each bundle, both selected-minus-control contrasts share the exact selected model and evaluation contract. Crossing memberships with bundles separates this comparison from one where both the selected set and reference are redrawn. The implemented selector uses the Superfiltering-style (SF)

score under our saved rendering; the experiment is not an exact reproduction of the full original Superfiltering training recipe.

The crossed-control test uses Qwen2.5-7B-Instruct (Qwen et al., 2024); the execution test uses Mistral-7B-Instruct-v0.3 (Jiang et al., 2023; Mistral AI, 2024). NF4 QLoRA uses rank 16, learning rate 2×10^{-4}, and 1,478,701 actual loss-target tokens per focal fit (Dettmers et al., 2023). Two RNG bundles vary model randomness and example order together. Equal actual loss-target tokens specify the training budget, rather than equal updates, wall time, or repetitions of every example. The same memberships link the comparator and support studies, although the focal results use different model families. Appendix A records the configuration.

3.2 Outcomes and the Paired Execution Interface

The broad evaluation is an equal-weight mean of IFEval, MMLU, BBH, GSM8K, and HumanEval benchmark scores on 2,050 items (Zhou et al., 2023b; Hendrycks et al., 2021; Suzgun et al., 2023; Cobbe et al., 2021; Chen et al., 2021). Task prompts, limits, and greedy decoding are fixed. The macro averages benchmark scores, not pooled items; IFEval uses prompt-level strict accuracy.

For execution, we identify the first eligible program/output boundary in each model's generated response. Replay supplies its generated output; execute supplies its sandbox execution result. Each pair shares the preceding context and program and receives the same continuation-token allowance, computed from the original replay branch. Supplied feedback strings may have different token lengths. Eligible numeric scalars are normalized; other feedback follows the fixed raw policy. Questions without an eligible boundary contribute identical branch scores. The intervention runs the model's own program rather than substituting a correct reference program. Replay is a controlled continuation after supplied feedback; its score need not equal the unmodified original completion's score. For active pairs, correctness is assessed only on the newly generated continuation after the supplied feedback; the feedback itself and the preceding text are excluded from the scored response (Appendix G).

Pairing occurs within each model–question trajectory. Different models need not produce the same prefix, code, or intervention opportunity. Within a pair, however, replay and execute differ at the feedback supplied after the same generated program, and subsequent continuation may change. This local design avoids attributing the effect to a wholly different prompting strategy or a fresh solution generated from the start. It also allows execution to return an error or an unhelpful result: successful program execution and a correct final answer are distinct outcomes.

The focal GSM8K cohort contains 128 questions; the reported SVAMP cohort contains 192 (Patel et al., 2021). Scores average two fixed fits per arm, retaining every question in the denominator. Model-dependent eligibility therefore remains part of the measured effect.

Uncertainty. Feedback intervals use 20,000 shared-question bootstrap replicates across arms and both fixed fits, averaging the fits within each replicate. Qwen broad-evaluation intervals instead use a paired, stratified normal approximation: benchmarks receive equal weight, with equal subject or subtask weights within MMLU and BBH. Both calculations condition on the fitted models and realized datasets (Dror et al., 2018; Bouthillier et al., 2021). For answer sensitivity, unresolved labels vary over their allowed scores, and bootstrap percentiles of the signed contrast bounds form a label/question envelope. Appendix I specifies both interval calculations and the label bounds. The envelope addresses the declared ambiguities, not undetected annotation errors.

4 Results: From Reference Choice to Execution

4.1 Accepted References Reverse the Point Ranking

The first Qwen bundle scores $64.39/68.31/70.96$ for $A/S/B$; the second scores $64.76/69.81/70.25$. The selected-minus-A differences are $+3.92/+5.04$ pp, whereas selected-minus-B differences are $-2.65/-0.44$ pp. Thus $B > S > A$ in both macro point rankings (Figure 1, left). The ordering is not shared by every benchmark; Appendix J.1 reports the constituent scores.

Replacing the comparator moves the reported advantage by -6.57 pp and -5.49 pp. The corresponding $B - A$ intervals are $[3.97, 9.17]$ and $[2.92, 8.05]$. In each bundle, the change exceeds the selected-over-A gain. Both references pass the declared balance criteria, and the selected fit is shared, so neither a rejected matching proposal nor

Gap	Replay	Execute	Reduction
$S - A$	$+10.94$	-1.95	12.89
$S - B$	$+2.34$	-0.78	3.13

Table 2: **Reference choice changes support dependence.** Gaps and reductions in percentage points on the same 128 GSM8K questions, averaging two fixed fits. These are point estimates; the intervals for I_A, I_B, and their difference H are stated in the text.

Arm	Eligible	Repair	Harm	Net	Gain (pp)
A	146	39	1	38	14.84
S	107	12	7	5	1.95
B	53	13	0	13	5.08

Table 3: **Execution changes the models differently.** Frozen numeric-score transitions on GSM8K. Counts are over 256 fit–question records per arm. Eligible records include unchanged and unsuccessful continuations; gain uses the full denominator.

a change to the selected model explains the difference between these comparisons. The second $S - B$ interval includes zero: the finding is two point-rank reversals and substantial control differences, rather than two independently significant negative selection effects.

A task-specific bridge to the execution study. The earlier corrected Mistral evaluation also ranks $B > S > A$ on GSM8K in both fits. Its mean gaps are $S - A = +13.67$ pp and $S - B = -2.15$ pp; $B - A = 15.82$ pp with a saved item interval $[10.77, 20.87]$. These are the same six $S/A/B$ adapters subsequently used for the paired execution study. This connects the reference question to the models being diagnosed, although it is a task-specific result, not a claim about their broad macro ranking. The earlier original-completion evaluation uses 256 questions; the focal replay/execute cohort uses 128 different questions and its own scoring protocol. We retain them as separate analyses (Appendix D).

4.2 The Comparator Changes the Execution Diagnostic

Under replay, Mistral S scores 36.72%, A 25.78%, and B 34.38%. Under execution, they score 38.67%, 40.63%, and 39.45%, respectively. The resulting comparison is a reference-by-support crossing on the same models and questions (Table 2).

Against A, the selected model goes from ahead to behind: $+10.94$ to -1.95 pp. Control A gains 14.84 pp from execution, compared with 1.95 pp for S, giving $I_A = 12.89$ pp and a 95% interval of $[7.81, 18.36]$. This interval establishes a change in the gap, not a significant negative final gap.

Against B, the gap changes from $+2.34$ to -0.78 pp. Because B gains 5.08 pp, the reduction is $I_B = 3.13$ pp, with interval $[-1.17, 7.81]$. We therefore do not treat the second line as another independently established interaction. Instead, the direct paired comparison between the two reductions is $H = 9.77$ pp, with interval $[4.69, 14.84]$. The comparison is made directly, not inferred from one interval excluding zero and another including it.

The reference consequently changes both the initial advantage and the amount by which execution reduces it. This is the empirical connection between the two studies: accepted controls need not provide the same performance reference or the same response to computational support. All six Mistral models, the questions, and the selected-model continuation pairs are shared across the two reference contrasts. Different models can still generate different programs and intervention opportunities.

4.3 Repairs and Harms Account for the Change

Table 3 expresses the frozen-score effect as changes in final correctness. Each arm has 256 fit–question records: 128 questions under two fixed fits. Execution repairs 39 errors for A and introduces one, yielding 38 additional correct records. For S, 12 repairs are offset by seven harms, yielding five. Their difference is 33 net successes over 256 records, corresponding to 12.89 pp. Control B has 13 repairs and no harms, yielding 5.08 pp; the 25-record difference between A and B is $H = 9.77$ pp.

The same support operation encounters different opportunities: A, S, and B have 146, 107, and 53 eligible records. Comparing success rates only among eligible responses would change the population differently for each arm. We retain the common full denominator, allowing both the availability of a program/output boundary and the subsequent continuation to contribute to the measured effect. The accounting explains where the observed score changes occur; it does not identify which training examples caused those behaviors. Appendix J reports the fit-level absolute scores and separates intervention availability, execution suc-

cess, and scalar-normalization eligibility.

Support across fits and question cohorts. The focal I_A is positive in both Mistral fits, $16.41\,\mathrm{pp}$ and $9.38\,\mathrm{pp}$. An earlier, disjoint 256-question GSM8K cohort, using the same four S/A adapters and normalized-feedback code, gives $13.09\,\mathrm{pp}$ with its saved conditional 95% interval $[9.18, 17.38]$. Its fit-specific values are $12.89/13.28\,\mathrm{pp}$. These are separate question-cohort checks on reused models, not independent training replications; we do not pool them. The focal H is also positive in both fits, $17.19/2.34\,\mathrm{pp}$, but their different magnitudes limit any claim of a stable effect over training randomness. Appendix C gives the resampling definitions.

4.4 Answer Interpretation Preserves Both Contrasts

We assess sensitivity to answer interpretation through a post hoc, unblinded AI-assisted review of saved responses, with author verification of 480 responses from the 240 focal pairs with unequal continuation text. The review distinguishes completed answers from incidental numbers, trace output, and unfinished calculations. Determinate corrections are applied in the sensitivity analysis; unresolved judgments define bounds. Appendix H specifies the review responsibility, coverage, policy, and examples; Appendix I gives the aggregation and uncertainty calculations.

After review, $I_A = 11.72\,\mathrm{pp}$ with a combined label/question envelope of $[6.25, 17.58]$, and $H = 8.59\,\mathrm{pp}$ with envelope $[3.52, 14.45]$. Both remain positive under the declared answer sensitivity (Figure 2). These envelopes combine unresolved-label choices with question resampling; they are not ordinary confidence intervals. Appendix I.2 gives the separate intervals and label ranges. The review supports the two contrasts, not a new absolute semantic ranking of the models.

Appendix E reports a supplementary A/B response comparison on SVAMP, without an S arm.

5 What the Comparisons Establish

The results identify two dependencies of a selection advantage: which accepted reference defines it, and how that reference responds to computational support. The Qwen reference change exceeds the original selected-over-A gain. On Mistral, the reference also changes the reduction attributed to execution. A favorable matched residual therefore describes a relation among data, reference, and evaluation interface.

Matching leaves behavioral variation. Passing the same balance checks does not make controls interchangeable. Source counts, sparse task labels, and token totals constrain observed composition while leaving content and the resulting response policies free to differ. The observed changes arise between accepted controls; they do not indicate that the balance checks failed to enforce their stated constraints.

Support gains require a specified comparison. A gain measured for one control cannot serve as a universal correction to a selected-data result. Nor does a larger execution gain identify the best final model: absolute scores, gains, and gap reductions answer different questions. For deployment with an interpreter, evaluation should report the reference, support interface, and repair–harm counts together. The measured operation includes model-dependent intervention availability and feedback rendering, so it does not isolate latent reasoning quality or a training mediator.

Evidence for realized comparisons. Point reversals make the instability visible; direct control differences and paired gap changes quantify it. The second Qwen $S - B$ interval includes zero, and the Mistral I_A interval does not establish a negative final gap. The direct H contrast nevertheless remains positive under the reported question analysis and answer sensitivity. These retrospectively selected comparisons provide evidence for the realized fits, not an estimate of reversal frequency across new pools, control draws, or training runs. Their contribution is the measured size and structure of reference sensitivity, rather than the subtraction identities defining it.

6 Related Work

Selection signals and composition. Likelihood-based IFD, Superfiltering, and T-SHIRT, together with quality/diversity and influence-based DEITA and LESS, motivate instruction-data selection (Li et al., 2024b,a; Fu et al., 2025; Liu et al., 2024; Xia et al., 2024). Length, style, and step-length studies show why a selection score and downstream utility must be distinguished (Zhao et al., 2024; Li et al., 2025b; Wang et al., 2026a). Mixture optimization and source-aware or stratified selection address composition (Renduchintala et al., 2024; Li et al.,

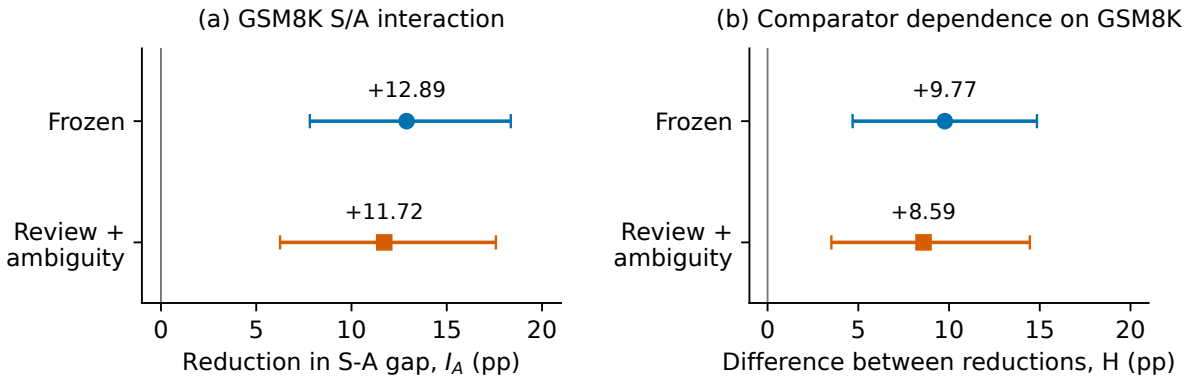


Figure 2: **The focal diagnostic and its comparator dependence survive the declared answer sensitivity.** Left: I_A, the reduction in the selected-minus-A gap. Right: $H = I_A - I_B$, the difference between the reductions under the two references on the same GSM8K cohort. Frozen estimates use 95% shared-question intervals. Review-reference estimates use combined label/question envelopes, which are not ordinary confidence intervals.

2025a; Mirza et al., 2025; Wang et al., 2026b); CommonIT and data-arrangement research also distinguish membership from presentation (Rao et al., 2024; He et al., 2025). We use composition to constrain reference sets, rather than propose a new selector or isolate an ordering mechanism.

Closest selection comparisons. Chasing Random and Rethinking Data Selection at Scale show that selection gains can be unstable across data and evaluation settings (Diddee and Ippolito, 2025; Xia et al., 2025). Controlled audits further separate representations from selection algorithms and identify budget-dependent gains (Nayak et al., 2026). Our question is narrower: can two controls satisfying the same declared balance constraints assign opposite point rankings to one selected fit? The crossed comparisons quantify this dependence without conflating comparator replacement with a change in the selected model.

Interfaces, execution, and measurement. Output-format research demonstrates that evaluation interfaces can change apparent training effects (Gan et al., 2026). Our added evidence is comparator-dependent support sensitivity: the two accepted references yield different execution-induced reductions on a common cohort. PAL, TATA, and tool-integrated training recipes study execution as a reasoning aid or learned policy (Gao et al., 2023; Xu et al., 2026; Cheng et al., 2026); here it is a local diagnostic of a selection contrast. Finding Answers in Thought and evaluation-harness guidance motivate checking answer extraction separately (Jo et al., 2025; Biderman et al., 2024). We therefore report frozen-score contrasts alongside sensitivity to the interpretation of saved answers, without treating the latter as independent gold labels.

Variability and reporting. Dodge et al. (2019) study development-compute dependence, and Bouthillier et al. (2021) distinguish sources of benchmark variability. The realized reference and its response to feedback add concrete comparison conditions to report. Our fixed-fit question intervals address neither variation over control draws nor new training runs.

7 Conclusion

Two references that satisfy the same composition and token constraints need not be interchangeable in measuring a selection advantage. Replacing the reference reverses the Qwen point ranking in both training bundles. On a common Mistral GSM8K cohort, execution reduces the selected-minus-A gap by $12.89\,\mathrm{pp}$ but the selected-minus-B gap by $3.13\,\mathrm{pp}$; the $9.77\,\mathrm{pp}$ difference links comparator choice to support dependence. The focal reduction and its comparator dependence survive the declared answer sensitivity. The point reversals are the visible consequence; the crossed comparisons, paired continuations, and repair–harm accounting establish the empirical argument. A favorable gap describes selected data in relation to a realized reference and support interface.

Limitations

The displayed experiments concern 1,000-example subsets, two 7B instruction-model families, and two fixed training bundles. The Qwen broad test

and Mistral task diagnostic are separate studies. The Mistral A/B-by-support comparison crosses references and interfaces locally, not across every family and benchmark. Question intervals condition on the realized fits. One program/output intervention does not measure a multi-step tool agent.

The displayed contrasts were selected retrospectively from a broader exploratory study; the intervals do not adjust for that selection or multiple comparisons. The companion archives reproduce only the displayed analysis scope, not all completed experiments. Task-family metadata cover only 4.5% of each training subset. Answer sensitivity uses post hoc, unblinded AI-assisted judgments and addresses recorded ambiguities, not unknown annotation errors. These fixed-fit comparisons do not establish universal reversals or cross-task robustness. The full historical computational cost and complete end-to-end GPU reproduction are not established.

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A Retained Configuration Details

Displayed percentages and contrasts are rounded independently; subtracting printed scores may differ by 0.01 percentage points. Training uses a frozen NF4 backbone with bfloat16 computation, LoRA rank 16, alpha 32, dropout 0.05, and no LoRA bias. Adapters target attention query/key/value/output projections and feed-forward gate/up/down projections. We use AdamW with learning rate 2×10^{-4}. Its coefficients are $(0.9, 0.999)$, epsilon 10^{-8}, weight decay zero, and maximum gradient norm one. Each forward/backward pass processes one example; gradients accumulate over up to four examples per optimizer step, with a potentially shorter final step. Linear warmup covers the first 3% of the target-token budget, followed by a constant learning rate, and the focal budget is 1,478,701 actual loss-target tokens. Maximum sequence lengths are 2,048 for Qwen and 4,096 for Mistral.

Token-weighted loss and exact-budget stopping. The focal $S/A/B$ fits use assistant-target causal cross-entropy, including the assistant end-of-sequence token when it is within the retained target budget. Prompt tokens do not contribute to the loss.

Let optimizer step t contain up to four single-example microbatches. If microbatch j contains n_{tj} unmasked target tokens and has mean target

loss $\bar{\ell}_{tj}$, the accumulated objective is

$$L_t = \sum_{j \in \mathcal{B}_t} \frac{n_{tj}}{\sum_{k \in \mathcal{B}_t} n_{tk}} \bar{\ell}_{tj}. \tag{5}$$

Thus, the step objective is a mean over its actual target tokens, not an equally weighted mean of example losses. Selection scores and source categories introduce no additional loss multipliers in these focal fits.

Training cycles through deterministically hash-ordered epochs until exactly 1,478,701 loss-bearing target tokens have been consumed. The ordering depends on the saved data-order seed, sampler seed, epoch, and example identifier. Every unique training example is exposed at least once.

If only r target tokens remain in the budget when the last example is reached, only its first r target labels contribute to the loss. The full input sequence through the assistant end-of-sequence token is retained, while excess target labels are set to the ignore value -100. The final optimizer step may contain fewer than four microbatches; its weights are normalized by the actual retained target-token count.

This rule fixes the total number of loss-bearing tokens. It does not equalize optimizer-step counts, repeated-example exposures, or source-specific exposure over the full run.

The first RNG bundle uses model seed 140181213, data-order seed 969304126, and sampler seed 952679015. The second uses 2692829214, 2403137412, and 196242321. Evaluation uses generation seed 65421376. A bundle therefore changes initialization-side randomness and example ordering together.

The broad contract contains 256 IFEval items, 1,140 MMLU items, 270 BBH items, 256 GSM8K items, and 128 HumanEval items. MMLU uses five-shot conditional likelihood sums; BBH uses three-shot chain-of-thought exact match and equal weighting over its 27 subtasks. GSM8K uses flexible extraction. HumanEval uses a fixed assistant prefill and one greedy sample. The feedback cohorts use numeric-answer scoring and preserve original records separately from the review sensitivity.

Execution occurs in a restricted sandbox. The feedback cap is 4,096 characters, the original generation cap is 1,024 new tokens, and paired branches use the saved remaining-token allowance. Execution may return an error: the intervention runs the model's program rather than substituting a reference solution. The packaged numerical records identify the displayed contrasts and supplementary checks; the plotting script regenerates the two figures. The package recomputes numerical analyses from saved records and reconstructs selection and matched memberships. It includes fixed training plans, metadata, and a training runner. Training text and reconstructable token payloads are withheld from the review archive; their original file hashes and a local restoration procedure are supplied. The restoration helper requires separately obtained exact historical inputs; it does not reconstruct them automatically from upstream pointers. Complete portable training, inference, and sandbox reproduction has not been verified.

B Realized Matched-Control Checks

The checks are applied separately to source and task-family distributions. Jensen–Shannon divergence uses natural logarithms; mass coverage is the selected-set probability mass of categories represented in the control. Single-pass token totals count every distinct training example once, not distinct vocabulary tokens. The original matching receipts record source Jensen–Shannon divergence and total variation of zero for both A and B. Task-family JS is 1.472×10^{-5} for A and 1.397×10^{-5} for B, with TV .001 and mass coverage one for each. The selected set contains 592,395 Qwen-rendered target tokens, counting each of the 1,000 distinct examples once; A/B contain 582,713/581,067. The accepted proposals are attempt two for A and attempt one for B. The controls share 193 examples with one another, while both exclude S under the declared identity rules. They are two fixed accepted realizations, not a sample sufficient to estimate the population distribution of all possible controls.

Matching terminology outside the focal comparison. The broader study also used an M2 specification that added constraints on input/output length, enumerated-step counts, mean enumerated-step length, response lines, and a categorical format signature. Those results are outside this manuscript's scope. Both focal controls use M1; B is another realization, not a more restrictive matcher.

C Question-Cohort Support for the Focal Interaction

The earlier cohort's saved 95% interval is $[9.18, 17.38]$, from 10,000 paired-question boot-

Cohort	N	Fit 1	Fit 2	Mean
Earlier	256	12.89	13.28	13.09
Focal	128	16.41	9.38	12.89

Table 4: Frozen numeric-score I_A in percentage points, using normalized feedback. Both cohorts reuse the same two S and two A Mistral adapters. Questions are disjoint; fits are not new replications.

strap replicates (seed 52001). The focal interval is $[7.81, 18.36]$, from 20,000 shared-question replicates (seed $20261009+128$). Both average the two fixed fits inside each replicate. We preserve the two saved analyses rather than pooling their estimates or representing them as prospective independent model confirmation. Adapter hashes match across the cohorts, and the focal construction excludes the earlier questions. The focal review-reference interactions are $14.84/8.59$ pp for the two fits; their unresolved-label point ranges are $[14.84, 15.63]$ and $[7.81, 9.38]$. These ranges address declared answer ambiguity and are not confidence intervals over training randomness.

D Earlier Mistral Task Comparison

The corrected original-completion evaluation contains 256 GSM8K questions. For the first fit, $A/S/B$ accuracy is $27.73/42.19/44.53$; for the second it is $19.92/32.81/34.77$. Thus both point rankings are $B > S > A$. The fit-average contrasts are $S - A = 13.67$ pp (saved 95% item interval $[8.13, 19.22]$), $S - B = -2.15$ pp ($[-7.69, 3.39]$), and $B - A = 15.82$ pp ($[10.77, 20.87]$). These intervals condition on the six fitted models and do not establish that B significantly outperforms S. The focal execution experiment reuses these adapters on its separate 128-question cohort. Original completions, controlled replay continuations, and their scoring contracts remain distinct; the earlier estimates are not pooled with the focal crossing.

E Supplementary Control Response on SVAMP

On the reported 192-question SVAMP cohort, A/B gain $11.46/4.95$ pp. The frozen-score contrast is $H = 6.51$ pp, with interval $[1.56, 11.46]$. This selected supporting comparison uses the same Mistral A/B fits and the common paired-feedback protocol. It measures differential control response, not a selected-versus-control reversal, because this cohort has no S arm. The central argument therefore rests on the GSM8K crossing. The displayed SVAMP result is a selected frozen-score view and does not establish robust transfer under all answer interpretations.

F Selection and Control-Construction Details

F.1 Scoring Context and Eligibility

The scoring input concatenates the system text, when present, a newline, the instruction, and another newline. The response is appended without rewriting. Tokenization uses `add_special_tokens=False`. The scorer is `openai-community/gpt2`, pinned to revision `607a30d783dfa663caf39e06633721c8d4cfcd7e`, with float32 parameters and eager attention.

The maximum scoring sequence length is 1,024 tokens. For an instruction of length L, at most $1{,}024 - L$ response tokens are retained. An example is eligible only if at least two response tokens remain and the response suffix obtained by tokenizing the concatenated instruction and response exactly matches the corresponding response-only token sequence. Instructions exhausting the available context, responses that are too short, and boundary mismatches are excluded before selection. The resulting scoring population contains 19,791 of the 20,000 reserved examples: 81 are excluded for prompt context, 115 for token-boundary mismatch, and 13 for too few response tokens. Scoring truncates 6,367 responses; this does not truncate their training targets. The same eligibility filter applies to the candidate roles; excluded examples are not replaced by additional draws.

Let u denote the instruction tokens and $r_{1:m}$ the retained response tokens. The conditional and unconditional losses follow the causal language-model label-shifting convention:

$$\ell_{\mathrm{c}} = -\frac{1}{m}\sum_{t=1}^{m} \log p_\theta(r_t \mid u, r_{<t}), \tag{6}$$

$$\ell_{\mathrm{u}} = -\frac{1}{m-1}\sum_{t=2}^{m} \log p_\theta(r_t \mid r_{<t}). \tag{7}$$

The unconditional loss excludes the first response token because no preceding token is supplied for its prediction. The clean IFD score is

$$\rho(u, r) = \frac{\exp(\ell_{\mathrm{c}})}{\exp(\ell_{\mathrm{u}})} = \exp(\ell_{\mathrm{c}} - \ell_{\mathrm{u}}). \tag{8}$$

Selection takes the 1,000 largest finite scores satisfying $0 < \rho < 1$, with example identifiers breaking

exact ties. Thus, the retained examples have the highest eligible ratios below one, rather than the lowest conditional loss.

Scoring truncation is recorded separately from training preprocessing. Selection does not replace the training response with its GPT-2 scoring prefix. Training instead uses the original response subject to the training tokenizer and sequence-length policy.

F.2 Calibrated Proposals and Acceptance

For source category c, let n_c be its count in S. The proposal mechanism preserves these source quotas. Its remaining calibration features comprise the number of target tokens, divided by the selected set's mean target-token count, and indicators for the observed task-family categories. Features are centered at their selected-set means.

For eligible candidate i in source category $c(i)$, the calibrated inclusion probability has the form

$$p_i = \sigma\left(\alpha_{c(i)} + \widetilde{x}_i^\top \beta\right), \tag{9}$$

where σ is the logistic function. Source-specific intercepts satisfy

$$\sum_{i:c(i)=c} p_i = n_c. \tag{10}$$

When a source has exactly as many eligible candidates as its quota, all its candidates receive inclusion probability one.

The calibration uses the maximum-entropy dual objective with quadratic regularization 10^{-9}. The implementation uses L-BFGS-B, at most 150 iterations, and coefficient bounds $[-32, 32]$. Randomized systematic rounding, with randomized ordering and shifts within source categories, converts the calibrated probabilities into subsets with exact source counts.

Calibration determines proposals; the realized balance checks determine acceptance. A proposal is retained only if its actual token total, categorical divergence, coverage, size, and exclusion checks satisfy the declared specification. Optimizer convergence alone does not establish acceptance. The first accepted proposal among at most 64 attempts is used.

This distinction matters because matching the expected features of a proposal distribution is not the same as matching the features of its realized subset. The checks in Appendix B concern the accepted datasets actually used for training.

F.3 Dataset Snapshot and Candidate Reservation

The general-instruction data originate from the training split of `allenai/tulu-3-sft-mixture`, pinned to revision `b14afda60f1bbebe55d5d2fa1e4df5042f97f8be`. All 20,000 rows in the focal reservation retain this raw dataset and revision in their provenance fields.

The 100,000-example parent is a derived, frozen research pool rather than the unmodified upstream training split. Its construction applies the saved source-eligibility and English-interaction rules, retains one eligible canonical example per lexical duplicate component, and excludes reserved development and calibration components. It also requires the full Qwen-rendered assistant target to fit within the 2,048-token training context. Overlong responses are excluded at this stage rather than truncated to manufacture eligible examples. Source proportions follow the post-filter population under the saved hash ordering; they are not equalized across sources.

After these exclusions, the first 100,000 rows under the saved ordering form the parent feature artifact `general_100000_features.parquet`. Its SHA-256 is `a93c4c7a410c329f1d25098a5228b5035f3a875ddd4b8ec06f5dc251c8d67c4c`.

The focal 20,000-example reservation is drawn from this parent after excluding examples, question groups, and duplicate components present in either of the two earlier 20,000-example research pools. Remaining candidates are ordered by the SHA-256 of the canonical JSON representation of the namespace `NLL_INDEPENDENT38_DISJOINT_SAMPLE_V1` and the duplicate-component identifier. The first 20,000 candidates are retained. This is a deterministic hash-based reservation, not a reservation selected using downstream benchmark scores. The subsequent scorer-eligibility filter leaves 19,791 examples without replacement or refill.

Reconstruction boundary. The CPU replay starts from the distributed frozen parent feature table and its recorded example, group, and duplicate-component identifiers. It reconstructs the focal reservation and memberships, but does not rerun the upstream English-interaction filter or lexical clustering from raw Tülu text. The source revision and row pointers support traceability; the review archive does not establish a complete raw-to-parent reconstruction.

F.4 Category Definitions and Realized Composition

The source category is the component-dataset identifier recorded in the Tülu mixture. The observed task-family category uses available source metadata: the recorded category field where available, and a frozen task-to-family mapping for documented SciRIFF tasks. Missing or conflicting metadata remain `UNKNOWN`. No new semantic classifier is used to fill missing labels.

Table 5 reports the source composition of the reservation and the three focal training sets. Each source count is identical across S, A, and B. The reservation counts precede scorer-eligibility filtering.

Only 45 examples per training set have an observed task-family label. Consequently, M1 provides exact source balance and close balance of the available task metadata, but does not establish that all latent task types or reasoning skills are balanced.

F.5 Why These Controls Enter the Focal Analysis

Control generation and selection of analyses for the manuscript are distinct stages. Control A is the first accepted M1 realization for the SF-selected set in the initial candidate-pool experiment. The positive matched residual observed in that experiment motivated a follow-up that varied the control draw and the training-randomness bundle separately.

Control B was generated as an additional realization using the stored calibration, unchanged matching thresholds, and a new sampling stream. Candidate proposals were accepted using balance and exclusion checks, not model performance. The realized draw seeds are 3205101666 for A and 703239051 for B; their accepted proposal indices are two and one, respectively.

After replacing A with B changed the point ranking under the first training bundle, a further experiment completed the missing B fit under the second bundle. It reused the exact B membership and the existing selected fit, with no new control search. The evaluation questions were also reused. This extension was informed by the earlier results and is not an independent, outcome-blind confirmation.

The present manuscript focuses retrospectively on the SF $S/A/B$ comparison from the broader exploratory study. These memberships and their Mistral counterparts provide linked comparator-replacement and execution-response analyses. This focus was chosen after inspecting prior results. The analysis therefore characterizes the displayed realized comparisons; it does not estimate how frequently the same pattern would occur among all acceptable controls.

G Exact Paired-Continuation Protocol

G.1 Initial Completion and Intervention Boundary

The initial completion uses the model's native chat template with a single user message:

```
Q: question
A: Let's think step by step.
```

No few-shot demonstrations are added. Generation is greedy, with at most 1,024 new tokens. The fixed stop strings are `Q:`, `</s>`, and `<|im_end|>`.

The parser searches for the first fenced block labeled `python` or `py`, ignoring label case. The opening fence must be followed by a newline. The first closing fence terminates the program. A complete fenced `output` block must follow it, with only whitespace between the two blocks.

The parser does not bypass an incomplete or ineligible first program to intervene at a later program. The shared prefix extends through the opening of the adjacent output block. The generated program and all preceding text remain fixed. Replay supplies the model-generated output; execute supplies the result of running that same program.

A record is inactive if this boundary is absent or incomplete, the original output exceeds 4,096 characters, no continuation budget remains, or the initial context check fails. For an inactive record, both branches retain the entire original response and its score. Its paired difference is therefore zero, and the question remains in the evaluation denominator.

G.2 Continuation Budget and Feedback Rendering

Let P be the encoded original prompt, C the encoded prompt plus shared prefix, and R the encoded original output followed immediately by its closing fence. The continuation allowance is

$$b = 1{,}024 - \big(|C| + |R| - |P|\big). \qquad (11)$$

This allowance is computed from the original replay representation before scalar normalization. Both branches receive the same b. Consequently, the protocol equalizes the permitted number of new

Source category	Reservation	Each of S, A, B
ai2-adapt-dev/evol_codealpaca_heval_decontaminated	3,515	288
ai2-adapt-dev/no_robots_converted	363	45
ai2-adapt-dev/numinamath_tir_math_decontaminated	2,304	71
ai2-adapt-dev/oasst1_converted	24	5
ai2-adapt-dev/personahub_code_v2_34999	658	64
ai2-adapt-dev/personahub_ifdata_manual_seed_v3_29980	1,128	142
ai2-adapt-dev/personahub_math_v5_regen_149960	6,164	103
ai2-adapt-dev/tulu_v3.9_aya_100k	75	6
ai2-adapt-dev/tulu_v3.9_open_math_2_gsm8k_50k	1,457	1
ai2-adapt-dev/tulu_v3.9_personahub_math_interm_algebra_20k	847	33
ai2-adapt-dev/tulu_v3.9_sciriff_10k	131	0
ai2-adapt-dev/tulu_v3.9_wildchat_100k	991	237
allenai/tulu-3-sft-personas-math-grade	2,343	5
Total	20,000	1,000

Table 5: Realized source counts. The final column gives the count in each individual training set, not the combined count across S, A, and B.

Observed task family	S	A	B
Brainstorm	12	11	13
Generation	32	33	31
Open QA	1	1	1
UNKNOWN	955	955	955
Total	1,000	1,000	1,000

Table 6: Observed task-family counts in the focal sets. Categories describe available metadata, not inferred semantic task coverage.

continuation tokens; it does not require equal feedback lengths or equal total input-plus-output token counts.

Scalar normalization applies only when execution succeeds and both the original output and actual feedback are standalone finite decimal scalars. The parser allows signed values, decimal fractions, properly grouped thousands separators, and scientific notation. Normalization removes commas, expands scientific notation, removes redundant fractional zeros, and maps signed zero to `0`. It does not round values or consult reference answers. Non-scalar feedback follows the fixed raw policy.

For normalized scalars, each feedback value is followed by a newline and a closing fence. For non-scalar pairs, replay preserves the original output and its closing fence, while execute appends a newline and closing fence to the actual feedback. The common encoded prefix is identical across branches. Each final branch is checked against the 8,192-token context limit; a feedback-induced overflow raises an error rather than silently removing a question.

Branch order is determined by a hash of the case, question, and branch identifiers. When the two branches have identical input token sequences, the saved continuation is reused. Otherwise, each branch is generated greedily with the same remaining-token allowance.

G.3 Execution and Error Handling

Execution runs the model-generated program once in an isolated Python process. The sandbox uses a separate filesystem root, an unprivileged identity, and system-call restrictions that disable network access and process creation. The recorded limits are five CPU seconds, eight wall-clock seconds, and 1 GiB of memory. The execution environment includes Python 3.13, SymPy 1.14.0, and mpmath 1.3.0.

The feedback contains actual standard output. A failed execution also supplies the recorded error representation. Embedded triple-backtick sequences are escaped to avoid creating an unintended feedback boundary. Feedback exceeding 4,096 characters is truncated and receives an explicit truncation marker.

Program errors and timeouts remain intervention outcomes; they are not grounds for excluding an otherwise active pair. An infrastructure failure that prevents reliable sandbox initialization instead stops execution. The protocol does not substitute a reference program, repair the generated code, or execute additional programs produced during the continuation.

G.4 What Is Scored

For active pairs, scoring applies to the newly generated continuation after the supplied feedback. The earlier reasoning, program, and supplied feedback

are not concatenated back into the scored answer. For inactive pairs, scoring retains the original completion, as described above.

This distinction prevents a supplied numeric result from being counted automatically as the model's final answer. It also explains why controlled replay and the original completion are separate evaluation conditions.

H Answer Extraction and Interpretation

H.1 Frozen Numeric Scoring

The primary numeric score uses the saved `flexible-extract` chain from `lm_eval` 0.4.13. Its regular expression is

```
(-?[$0-9.,]{2,})
|(-?[0-9]+)
```

where the displayed line break is typographical: the two lines form one expression. The filter selects the last match (`group_select=-1`), followed by the `take_first` filter.

The saved metric normalization removes commas, dollar signs, the configured answer-prefix pattern, and a terminal period. The extracted prediction and reference are then parsed as finite decimal values. Exact numeric equality receives score one; an invalid or unequal prediction receives zero. The separately retained official score uses the frozen exact-match metric.

These operations specify an extraction-based outcome. They do not establish that the extracted number is the quantity the response explicitly adopts as its answer.

H.2 Answer-Adoption Policy

The additional interpretation analysis applies the following policy to saved responses:

1. Accept a completed scalar answer stated in ordinary prose, a final-answer marker, or a clearly designated numerical answer expression. The quantity and units must match the question; a box or particular wording is not required.

2. Do not treat program constants, list indices, variable names, tracebacks, or supplied feedback alone as a newly asserted answer. A fenced block by itself is insufficient.

3. Retain a completed answer followed by verification unless the response retracts or replaces it. A later truncated verification does not automatically invalidate an already completed answer.

4. Assign an unresolved label $U = \{0, 1\}$ when the response retracts its answer without resolving the task, contains conflicting final answers, or leaves answer adoption ambiguous.

5. Assign zero to a clearly incorrect scalar, unit, or requested quantity, and to a response without a completed answer.

6. Record question/reference inconsistencies separately. The primary analysis retains the frozen dataset reference.

The policy was written before the uniform review pass, after earlier unblinded inspections had revealed extraction problems. This is therefore a post hoc, unblinded sensitivity analysis, not independent gold annotation or a prospective validation of the answer policy.

H.3 Review Responsibility, Coverage, and Decisions

OpenAI Codex performed the initial post hoc, unblinded answer-adoption review using the policy above. The assistant inspected the questions, reference answers, saved continuations, and existing scores, and proposed response-level judgments with reasons for corrections and unresolved cases.

An author subsequently inspected all 480 responses in the focal comparison, confirmed the nine determinate corrections, and retained the four unresolved labels. Codex therefore provided the initial annotations, while the author performed the final verification of the decisions used in the analysis. Model identities, conditions, and existing scores were visible during review. This procedure is author verification of AI-assisted annotations, not an independent blinded human annotation study.

Within the focal Mistral $S/A/B$ comparison, the review covers 240 replay/execute pairs with unequal continuation text, comprising 480 responses. Equal-score pairs are included: review eligibility depends on unequal text, not on whether the frozen correctness score changes. The unequal-text pair counts are 116 for A, 73 for S, and 51 for B.

The saved annotations contain nine determinate corrections to individual branch scores and four unresolved branch labels. The analysis reads these recorded decisions and applies a fixed aggregation procedure. Determinate judgments replace

the corresponding frozen scores in the review analysis. For an unresolved label, the review-reference estimate retains the frozen score, while sensitivity bounds allow both zero and one. Thus, unresolved cases are not assigned a favorable value to strengthen the reported contrast. The original frozen-score results remain separately reported.

Identical-text continuations contribute zero paired difference under the declared answer-adoption policy. Changed-feedback cases with identical continuation text were also inspected for dependence on the supplied feedback. This cancellation supports the paired contrasts; it does not establish semantic correctness of every absolute-score record.

The review is a sensitivity analysis conditional on the recorded AI-assisted annotations verified by an author. Its bounds address the explicitly recorded ambiguities, not undetected annotation errors. Broader use of Codex is disclosed in Appendix K.1.

H.4 Examples from Saved Responses

The following records illustrate the policy rather than define additional evaluation rules.

A later incidental number. Record `P0074`, an execute continuation for the first B fit, states:

> The output 21 indicates that Dean has $21 left after buying the toys and receiving the extra $10.

The reference is 21. Frozen extraction gives score zero, whereas the answer-adoption policy gives one because 21 is explicitly assigned to the requested remaining amount. The later 10 describes money received, not a replacement answer.

An unfinished calculation. Record `P0039`, an execute continuation for the first A fit, ends in unresolved cupcake algebra and a repeated attempt to solve the constraints. Its last extracted number matches the reference value 3, but it does not complete an answer asserting that value. The frozen score of one is changed to zero.

Conflicting answers. Record `P0043`, an execute continuation for the first A fit, gives a boxed answer of -4 and then states that the person can miss up to four classes. The reference is 4. The disagreement between the boxed answer and the prose is retained as $U = \{0, 1\}$ rather than resolved in whichever direction strengthens the aggregate contrast.

I Estimands, Resampling, and Label Bounds

Let $y^e_{gsi} \in \{0, 1\}$ be correctness for arm g, fixed fit $s \in \{1, 2\}$, question i, and interface $e \in \{r, x\}$. Define the question-level, fit-averaged execution effect

$$z_i(g) = \frac{1}{2} \sum_{s=1}^{2} \left(y^x_{gsi} - y^r_{gsi} \right). \tag{12}$$

The reported gain in percentage points is

$$\widehat{\Delta}_g = \frac{100}{N} \sum_{i=1}^{N} z_i(g). \tag{13}$$

The question-level vectors for the gap reductions and their difference are

$$v_i^{I_A} = z_i(A) - z_i(S), \tag{14}$$

$$v_i^{I_B} = z_i(B) - z_i(S), \tag{15}$$

$$v_i^{H} = z_i(A) - z_i(B). \tag{16}$$

Thus, H is computed directly on paired observations. It is not obtained by comparing whether separate intervals exclude zero.

I.1 Shared-Question Bootstrap

For each replicate, draw question counts

$$(n_1^*, \ldots, n_N^*) \sim \text{Multinomial}\left(N; \frac{1}{N}, \ldots, \frac{1}{N} \right). \tag{17}$$

The same counts apply to every arm, interface, and fixed fit. For contrast vector v_i, the replicate estimate is

$$T^* = 100 \sum_{i=1}^{N} \frac{n_i^*}{N} v_i. \tag{18}$$

The focal analysis uses 20,000 replicates and NumPy's default random-number generator with seed $20261009 + N$. The 2.5th and 97.5th percentiles give the reported interval.

Fits are averaged within each question before resampling. Neither fits nor selected or control datasets are resampled. The resulting uncertainty concerns questions conditional on the realized datasets and fitted models. The earlier cohort retains the separately saved bootstrap specified in Appendix C. These intervals are descriptive and are not adjusted for selection across the broader sequence of analyses.

I.2 Unresolved-Label Bounds

Write a contrast as

$$T = \sum_j a_j y_j, \tag{19}$$

where j indexes distinct branch-score records and a_j includes the contrast sign, averaging weight, and percentage-point scaling. Repeated occurrences of the same score are combined before bounding. In particular, the shared selected-model term cancels from H.

For a determinate label, let $\ell_j = u_j = y_j$. For an unresolved binary label, let $[\ell_j, u_j] = [0, 1]$. Then

$$T_{\min} = \sum_{a_j \geq 0} a_j \ell_j + \sum_{a_j < 0} a_j u_j, \tag{20}$$

$$T_{\max} = \sum_{a_j \geq 0} a_j u_j + \sum_{a_j < 0} a_j \ell_j. \tag{21}$$

These are sensitivity bounds over the declared label choices.

For the combined label/question envelope, the bounds are recomputed under each shared-question bootstrap weighting. The reported lower endpoint is the 2.5th percentile of replicate lower bounds; the upper endpoint is the 97.5th percentile of replicate upper bounds. The envelope incorporates the specified answer ambiguities and question resampling. It does not estimate unknown annotation errors or variability over new training runs.

Separate answer-sensitivity summaries. For I_A, the review-reference estimate is $11.72\,\text{pp}$ with a conditional 95% question interval $[6.64, 16.80]$. Varying only unresolved labels gives the point range $[11.33, 12.50]$; combining label choices with question resampling gives the envelope $[6.25, 17.58]$. For H, the corresponding estimate is $8.59\,\text{pp}$, the conditional interval is $[3.52, 13.67]$, the unresolved-label point range is $[8.59, 9.38]$, and the combined envelope is $[3.52, 14.45]$. The intervals hold the review-reference labels fixed, whereas the envelopes also propagate their declared unresolved choices.

I.3 Broad-Evaluation Intervals for Qwen

The broad Qwen analysis uses approximate normal intervals computed from paired item contrasts, rather than the question bootstrap used for feedback analysis. Predictions are aligned by benchmark and item identifier. The implementation additionally checks agreement of document hashes, prompt hashes, and subject labels across the models entering a contrast.

For a fixed training bundle, let y_{mbhi} be the score of model m on item i in stratum h of benchmark b. Define the item-level linear contrast

$$d_{bhi} = \sum_m c_m y_{mbhi}. \tag{22}$$

For example, $S - A$ uses coefficients $+1$ for S and -1 for A. Model scores are combined on the same item before the variance is calculated, preserving their observed within-item covariance.

Each of the five benchmarks receives weight $1/5$. MMLU is further divided into 57 subject strata, each containing 20 items; BBH is divided into 27 subtask strata, each containing 10 items. IFEval, GSM8K, and HumanEval each constitute one stratum, containing 256, 256, and 128 items, respectively. Let H_b be the number of strata in benchmark b, and define

$$w_{bh} = \frac{1}{5 H_b}. \tag{23}$$

With n_{bh} items per stratum, the macro contrast and its estimated standard error in percentage points are

$$\widehat{T} = 100 \sum_{b=1}^{5} \sum_{h=1}^{H_b} w_{bh} \bar{d}_{bh}, \tag{24}$$

$$\widehat{\mathrm{SE}}(\widehat{T}) = 100 \sqrt{\sum_{b=1}^{5} \sum_{h=1}^{H_b} w_{bh}^2 \frac{s_{bh}^2}{n_{bh}}}, \tag{25}$$

where $\bar{d}_{bh} = n_{bh}^{-1} \sum_i d_{bhi}$ is the stratum mean and s_{bh}^2 is the sample variance of the paired item contrasts, calculated with denominator $n_{bh} - 1$. The displayed approximate 95% interval is

$$\widehat{T} \pm 1.96\, \widehat{\mathrm{SE}}(\widehat{T}). \tag{26}$$

Thus, benchmark weights enter the point estimate linearly and the variance estimate quadratically. Pooling all 2,050 items with equal weight would produce a different estimand and is not the implemented calculation.

For comparisons that reuse a fitted model, its coefficients are combined algebraically before constructing item contrasts. In particular, the selected-model term cancels when comparing the two selected-minus-control gaps. The uncertainty of that difference is computed from the remaining

Fit	Arm	Original	Replay	Execute	Gain
1	A	27.34	27.34	46.88	19.53
1	S	40.63	40.63	43.75	3.13
1	B	40.63	40.63	42.97	2.34
2	A	24.22	24.22	34.38	10.16
2	S	32.81	32.81	33.59	0.78
2	B	28.91	28.13	35.94	7.81

Table 7: Focal Mistral GSM8K scores by fixed fit. Accuracy columns are percentages; gain is execute minus replay in percentage points. Displayed values are rounded independently.

paired control scores, not by treating the two gap estimates as independent.

These intervals condition on the realized training datasets, fitted models, and evaluation strata. The calculation estimates within-stratum item variance and includes no cross-stratum covariance term. It does not estimate variability over new training seeds, control draws, candidate pools, or benchmark collections. The intervals are pointwise and do not adjust for the outcome-informed selection of analyses from the broader experimental sequence.

J Fit-Level Scores and Intervention Coverage

Table 7 gives the absolute frozen numeric scores underlying the focal crossing. "Original" denotes the initial completion; "Replay" and "Execute" denote the controlled branches. All six rows use the same 128 questions.

The fit-specific values of I_B are -0.78 and 7.03 percentage points. The corresponding values of H are 17.19 and 2.34 points. The direction of H is shared across the two fits, while its magnitude varies substantially.

Table 8 separates intervention availability, successful execution, scalar-normalization eligibility, and unequal continuation text. Successful execution is a program-level outcome, not a correctness judgment about the final answer.

For arm g, let a_{gsi} indicate an active intervention. Because inactive pairs have zero difference, the finite-sample gain admits the accounting identity

$$\widehat{\Delta}_g = 100\,\widehat{p}_g\,\overline{d}_{g,\text{active}}, \tag{27}$$

where $\widehat{p}_g$ is the active fraction of the $2N$ fit–question records and $\overline{d}_{g,\text{active}}$ is their mean paired correctness difference. This identity separates opportunity frequency from the net response within opportunities. It is descriptive: the active subsets

Record category	A	S	B
All fit–question records	256	256	256
Active intervention	146	107	53
Successful execution	126	98	35
Scalar-normalizable pair	101	55	14
Unequal continuation text	116	73	51

Table 8: Coverage of the focal paired protocol. Successful execution and scalar-normalizable pairs are subsets of active records. Unequal text need not imply a changed correctness score.

Fit	Benchmark	A	S	B
1	IFEval	60.55	59.77	63.67
1	MMLU	70.96	70.61	72.02
1	BBH	53.70	60.00	65.19
1	GSM8K	75.00	79.30	78.12
1	HumanEval	61.72	71.88	75.78
2	IFEval	57.42	63.67	60.94
2	MMLU	69.21	70.35	71.14
2	BBH	51.48	61.11	63.70
2	GSM8K	71.48	78.12	82.03
2	HumanEval	74.22	75.78	73.44

Table 9: Qwen benchmark scores (%) by fixed training bundle. IFEval uses prompt-level strict accuracy; MMLU and BBH retain equal subject/subtask weights. The macro gives each of the five benchmarks equal weight.

are generated by different models and are not a common randomized stratum. The primary comparisons therefore retain all questions.

J.1 Qwen Benchmark Breakdown

Table 9 decomposes the five-benchmark macro in Section 4.1. The macro ordering $B > S > A$ holds in both bundles, but the constituent rankings differ: $S > B$ on GSM8K in the first bundle and on IFEval and HumanEval in the second. These scores retain the original evaluation contract and do not add new fits or inference. The companion analysis emits them as `tables/qwen_benchmark_scores.csv`.

K Runtime and Study Lineage

Recorded feedback runtime. The focal feedback run records an NVIDIA A100-SXM4-40GB, Python 3.13.15, PyTorch 2.11.0 with CUDA 12.8, Transformers 4.57.3, PEFT 0.17.1, bitsandbytes 0.48.1, and Accelerate 1.10.1. Initial completions use batch size eight; paired continuations use batch size one. The saved configuration enables deterministic settings and disables TF32. These versions describe this feedback run, rather than asserting that every earlier training and scoring

stage used the same environment.

Pinned training-model snapshots. The Qwen checkpoint is `Qwen/Qwen2.5-7B-Instruct`, revision `a09a35458c702b33eeacc393d103063234e8bc28`

The Mistral checkpoint is `mistralai/Mistral-7B-Instruct-v0.3`, revision `c170c708c41dac9275d15a8fff4eca08d52bab71`

Training configuration and random-number bundles are specified in Appendix A.

Membership and model reuse. The Qwen comparator crossing uses three fixed memberships under two training bundles, giving six fitted models. The two comparator contrasts within a bundle share the same selected fit. The Mistral study trains six corresponding models using the same memberships. Its corrected original-completion analysis rescores saved predictions; it does not add training replications.

The earlier S/A execution cohort and the focal $S/A/B$ cohort reuse the corresponding Mistral adapters. Their question cohorts are disjoint, but their training histories are shared. The supplementary SVAMP comparison also reuses the A/B adapters. Additional question evaluations therefore expand the observed evaluation coverage without increasing the number of independent trained models.

Training-record coverage. All twelve focal training plans are included. The distributed exposure ledger covers all six Mistral fits but only the first Qwen S/A fits; the remaining four Qwen fits have plans without matching exposure-ledger entries in this release. Plan validation checks recorded membership, lengths, and target budgets, not independent execution of the historical GPU runs. The coverage is listed in `data/provenance/training_record_coverage.json`.

Saved-record traceability. The research records retain question identifiers, model and input checksums, intervention status, feedback normalization decisions, remaining-token budgets, branch responses, extracted predictions, and scores. These fields distinguish inactive passthrough records, identical-input reuse, failed program executions, and completed paired continuations. Numerical aggregation preserves the common question universe and the pairing of branches and models. The scope of the accompanying reproducibility package remains as stated in Appendix A.

Benchmark source revisions, splits, and deterministic selection rules are documented in `data/provenance/evaluation_sources.json`. The broader research had already evaluated all 1,319 GSM8K test questions on other fits; focal and earlier cohorts are disjoint under their declared exact/near-match rule, not globally unseen. A release audit compared 2,807 distinct training questions against 2,050 broad evaluation questions after NFKC, case, and whitespace normalization and found no full-string matches. This does not exclude paraphrases, partial overlap, or pretraining contamination.

The archives cover the focal comparisons described in this manuscript. The included/excluded analysis boundary is documented in `data/provenance/package_scope.json`; retained records inside each included cohort are not filtered by outcome. The whole historical computational budget is not recoverable from the supplied records.

Source-specific licenses and attribution accompany the archives. The Tülu collection declares ODC-BY-1.0 with component-specific conditions, including noncommercial material. Contact-like content of unverified authenticity was found in upstream training material. The distributed copy withholds training prose and reconstructable tokens, retaining IDs, lengths, original hashes and plans; a separate restoration helper verifies original payload hashes. Nine IFEval generated records similarly have text/token payloads withheld while scores remain unchanged. These documented distribution changes limit direct GPU and payload-level reproduction without changing experimental memberships or numerical results. Model weights and adapters are not included.

K.1 Use of AI Assistance

OpenAI Codex was used to assist with developing and refining experimental plans, implementing and debugging Colab notebooks and research pipelines, inspecting saved experimental outputs, implementing statistical analyses and figures, identifying relevant literature, and drafting and revising manuscript text and LaTeX. Its involvement included methodological discussion and interpretation of results, beyond language proofreading.